

diff - diff

**There's a clean line between
treated and control.**

Until there isn't.

Spillover-Aware DiD.

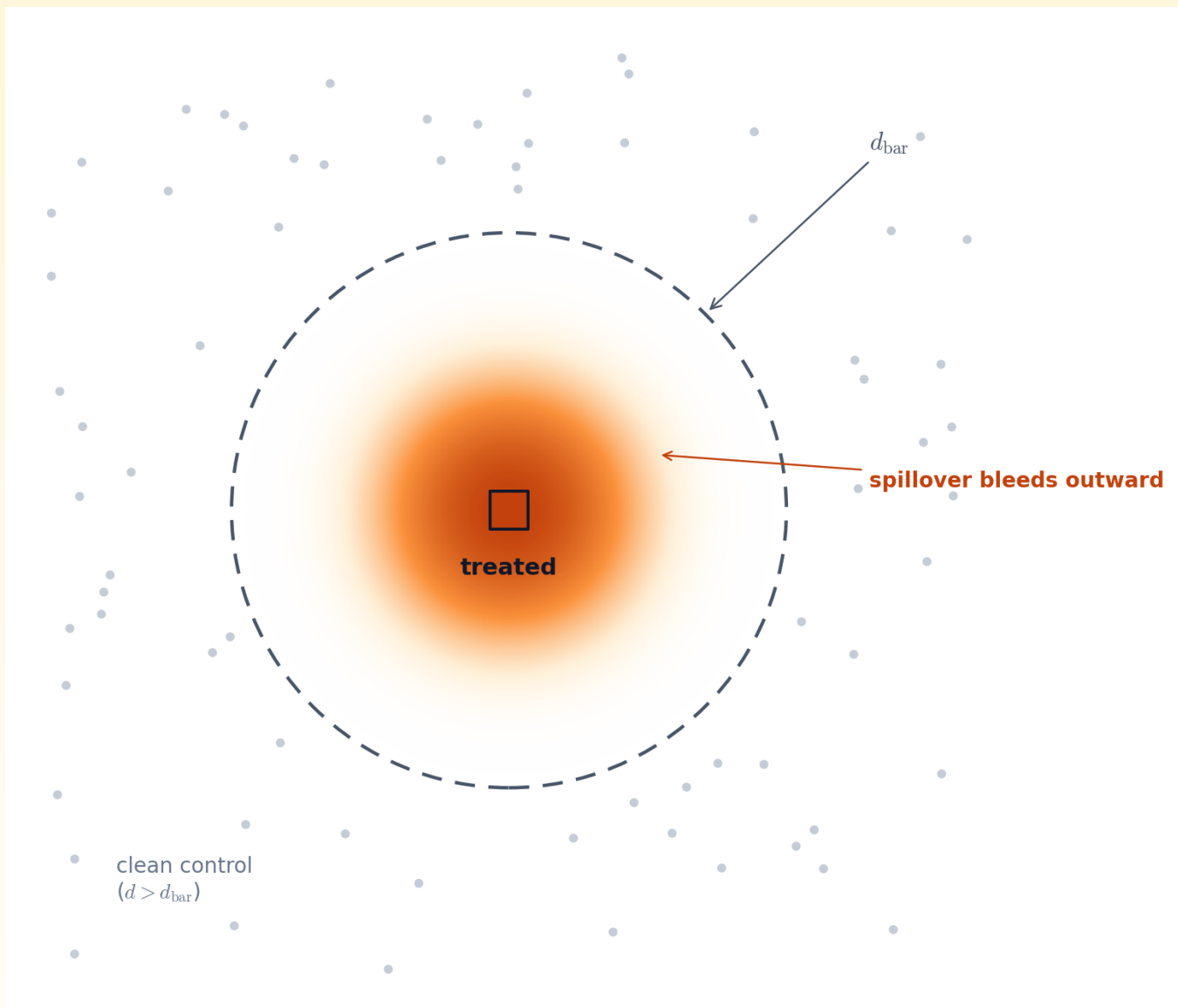
Now in diff-diff.

Butts (2021).

arXiv:2105.03737

diff-diff v3.4.1

Some treatment leaks into your controls.



The 'control' isn't. (Partly.)

Sometimes treatment **leaks** across borders.

Place-Based Economic Policy

Enterprise zones lift local investment - and bleed into adjacent neighborhoods.

Out-of-Home Media Buys

Billboards reach commuters who live outside the placement DMA.

Geo-Targeted Digital Campaigns

Meta / Google geo-fences leak via household-IP and travel boundaries.

Retail Footprint Tests

A new store lifts sales in adjacent trade areas, not just at the location.

The SpilloverDiD estimator.

For partial-contamination DiD.

Recovers both direct effects on treated
and per-ring spillover on near-controls.

Two effects, one regression.

Direct τ_{total} on treated AND per-ring δ_j on near-controls.

Far-away controls anchor identification.

Beyond d_{bar} , controls are uncontaminated.

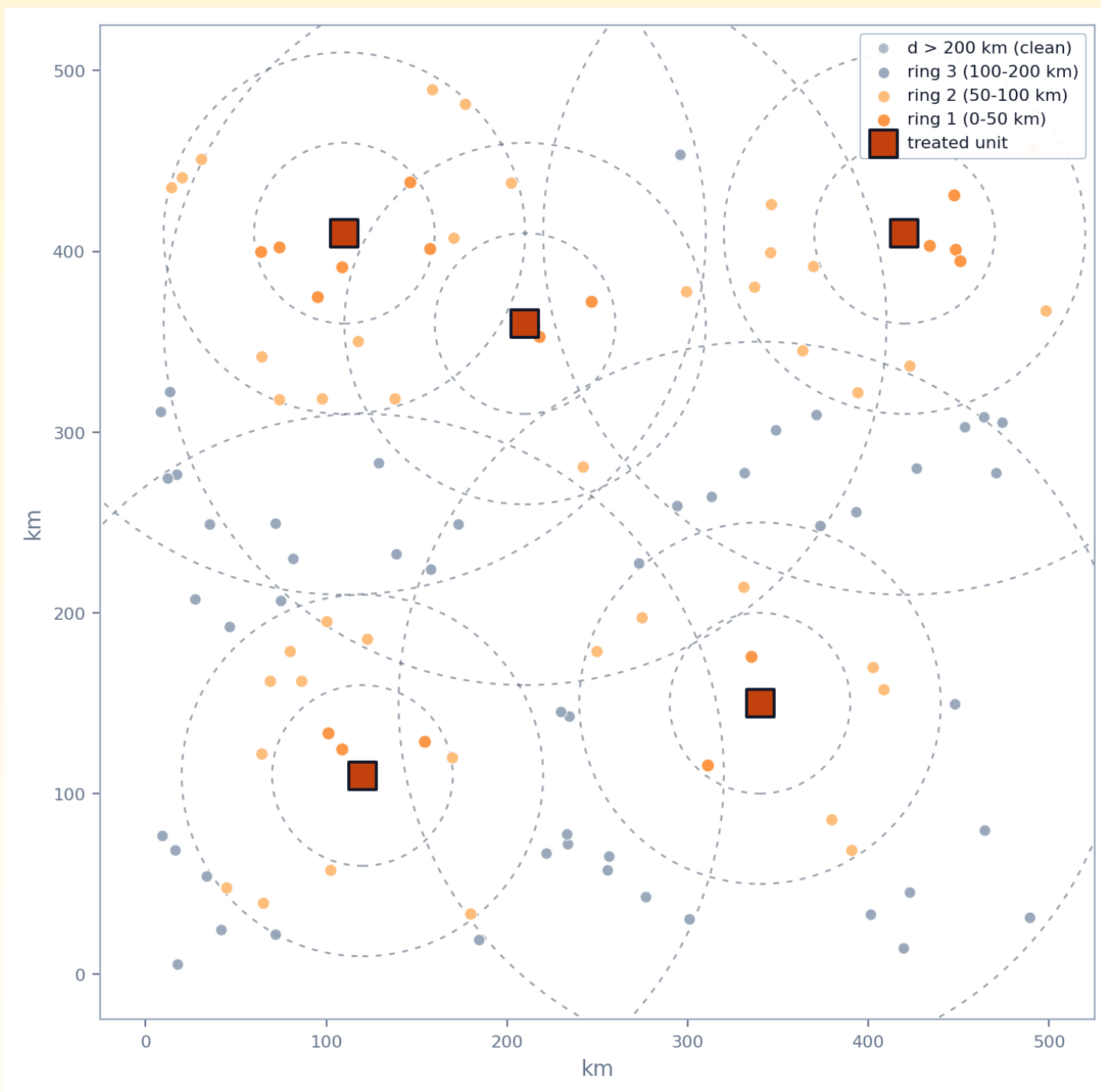
Spatial-HAC inference, out of the box.

Conley (1999) panel-block standard errors.

Built on two-stage Gardner (2022) DiD.

diff-diff v3.4.1

Identify the spillover ring by ring.



Each ring estimates its own spillover effect.

The hidden bias.

(Butts 2021, Proposition 2.1)

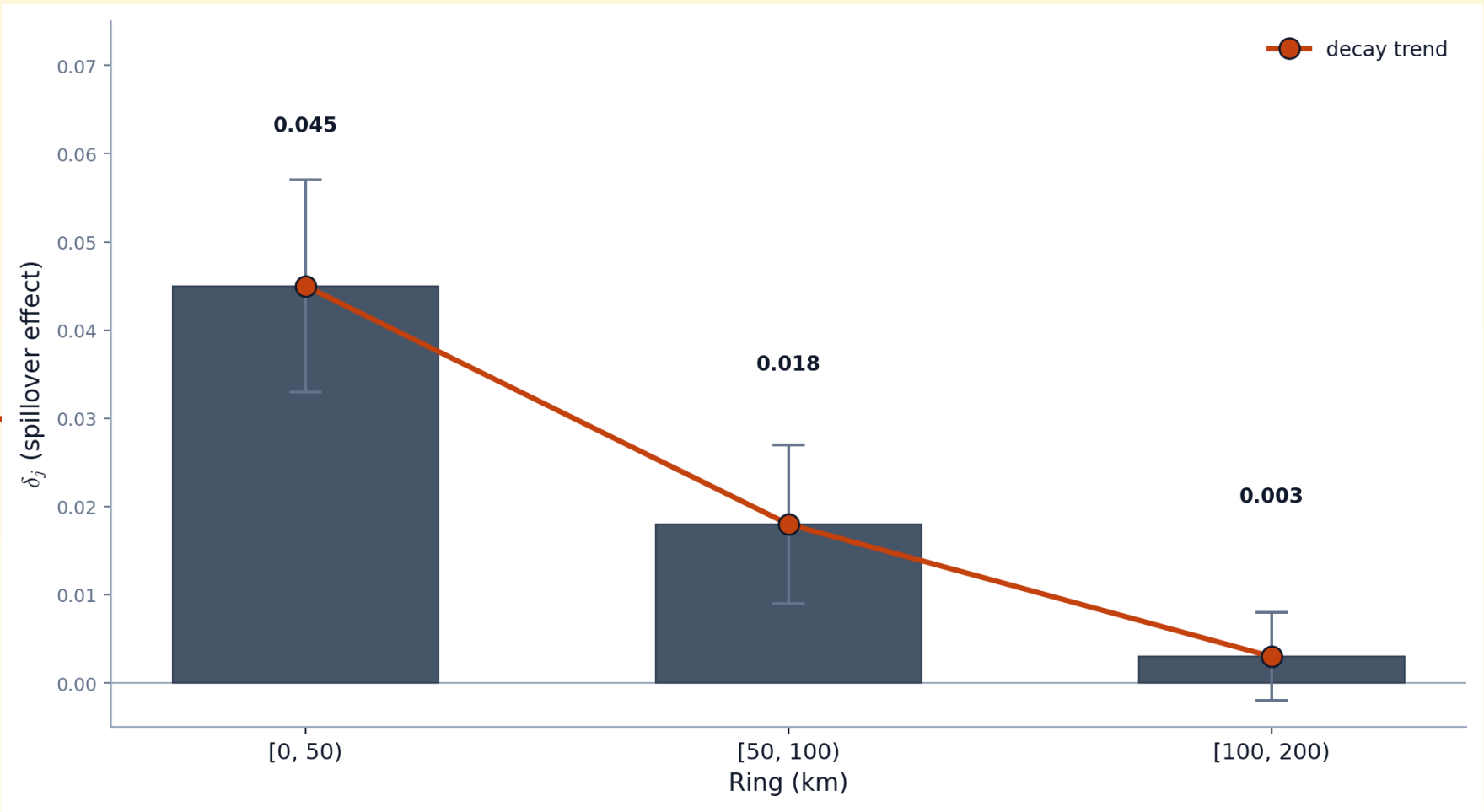
$$\beta_{\text{DiD}} \approx \tau_{\text{total}} - \tau_{\text{spill}}(0)$$

the hidden bias term

Standard DiD recovers τ_{total} only when spillover is zero.

SpilloverDiD identifies τ_{total} and δ_j separately.

Spillover decays with distance.



Per-ring δ_j attenuates toward zero.

The Code.

Same sklearn-like API as every diff-diff estimator.

```
from diff_diff import SpilloverDiD

result = SpilloverDiD(
    rings=[0, 50, 100, 200],
    conley_coords=('lat', 'lon'),
).fit(
    data,
    outcome='sales', unit='store',
    time='week', treatment='campaign',
)

print(result.att)          # tau_total = 0.124
print(result.spillover_effects) # per-ring delta_j

#           coef    se    p_value
# [0, 50)    0.045  0.012  0.001
# [50, 100)   0.018  0.009  0.041
# [100, 200)  0.003  0.005  0.561
```

Two outputs. One fit() call.

Production-ready.

Spatial-HAC SEs

Conley (1999) panel-block,
kdtree fast path

Cluster-Robust

HC1 / CR1 with
Gardner GMM correction

Survey Design

pweights, strata, PSU,
FPC via Binder TSL

Subpopulation Domains

Full-design retention
via zero-pad scores

Event-Study Mode

Per-event-time x ring
decomposition

Staggered Timing

Gardner two-stage;
non-staggered too

All composable on the basic fit() call.

Validated.

Paper equations + Monte Carlo coverage on synthetic DGPs.

Butts Eq 5/6 + Proposition 2.1

Ring-augmented spec + bias decomposition, implemented as written.

Gardner Two-Stage GMM (Section 4)

Stage-1 FE absorption + stage-2 with first-stage uncertainty correction.

Conley (1999) Spatial-HAC

Matched to R conleyreg at atol=1e-6 on parity fixtures.

Monte Carlo Coverage

Recovers known $\tau_{\text{total}} + \delta_j$; coverage near nominal level.

Now in diff-diff.

Spillover-Aware DiD.

```
$ pip install --upgrade diff-diff
```

github.com/igerber/diff-diff

diff - diff

Difference-in-Differences for Python